An Exploration of Isaac Levi's 'q' Parameter in Betting on Horse Races

Lucy Levenson

PHI 533: Decision Theory: Taking Risks in Epistemology

I. Introduction

This paper investigates the application of Isaac Levi's decision-making framework to hypothetical horse race betting settings. Levi's formula accommodates a decision-making agent's particular attitudes in the tradeoff between seeking knowledge and avoiding error and can thus provide insightful analysis of the effects of weighing these objectives differently. I will explore the application of Levi's models to cases involving informational imbalance or different interpretations of evidence and premiums based on decision specificity.

This is useful in various cases. In a case where all the potential utilities are outlined such as in a bet on horse racing, the optimal balance between seeking knowledge and avoiding error could be used to make the bets that most often maximize profit. In cases where all potential utilities are less distinct (i.e. not in the form of money or anything quantitative), estimations of the benefit of being specific can allow for some guidance in which conclusions are likely to be the most fruitful. This might look like an assessment of approximate utilities to determine whether a given situation should prioritize relief from agnosticism, avoidance or error, or strike some sort of balance.

I will begin by summarizing the decision-making framework used in this paper as described in Isaac Levi's 1967 book *Gambling with Truth*.[1] In developing this system, Levi

[1] Levi, Isaac. *Gambling with Truth: An Essay on Induction and the Aims of Science*. Alfred A. Knopf, 1967.

aimed to create a formula that weighs the goals of relief from agnosticism and avoiding error according to the preferences of the agent or the specific needs of the situation.

Gambling with Truth sets out a framework for rational decision-making that accounts for a variable preference between two goals: relief from agnosticism and avoiding error. Relief from agnosticism refers to the benefit one receives from narrowing down their decision problem. For example, in predicting the next day's weather, coming to the conclusion that it will rain provides more relief from agnosticism than coming to the conclusion that it will either rain or be sunny. To avoid error may be to not make a call on the weather forecast at all.

The ideal decision is both highly informative and correct, but these two goals work against each other, requiring the agent to make tradeoffs between them. An agent who seeks to maximize informativity will be inclined to make more specific predictions. Consider a case where specificity might be highly valuable: you are driving and there is a deer in the middle of the road. You must make a decision between slamming on the brakes, swerving left, or swerving right. Indecisiveness in this case is clearly the worst option, even at the cost of potentially choosing an imperfect option that might lead to a negative outcome such as swerving into a telephone pole. The driver is representative of the prioritization of relief from agnosticism, even when it exposes them to a higher potential for error.

An example of a situation that calls for an agent to maximize their avoidance of error at the sacrifice of their relief from agnosticism could be the dilemma faced by an adventurous foodie who wants to try pufferfish, a Japanese delicacy. The foodie is extremely curious about the taste and experience of eating pufferfish and would love to relieve their agnosticism. However, they are located in Princeton, New Jersey, and do not have access to pufferfish with a high probability of being prepared correctly. While they might want to relieve their agnosticism,

it is more important in this case that they prioritize avoiding error since the error of eating incorrectly prepared pufferfish could have extreme health consequences.

In general, seeking relief for agnosticism recommends specificity in decision-making, while seeking avoidance of error recommends vagueness. To improve in one area is to weaken in the other, yet they are both valuable goals. Levi establishes a formula for calculating the Epistemic Utility (EU) of a partition in a decision problem that depends on a parameter, 'q', which accounts for the agent's preference to either be more specific or more vague.

Levi's framework involves the calculation of EU for each partition of the decision problem and then the identification of the partition that maximizes EU. For a set of outcomes {X, Y, Z}, a partition can be any combination of the outcomes such as {X or Z}, {X}, or even {X or Y or Z}.

The EU of a partition 'H' is defined as the following: EU(H) = p(H) - q * cont(not-H)

The first term of the equation, 'p(H)', represents the goal of avoiding error. A less probable partition has a higher chance of error and its EU will be affected accordingly. However this may be outweighed by the second term of the equation, '- q * cont(not-H)', which represents the goal of relief from agnosticism. The 'cont' (content) of a partition indicates how informative it is. For a highly informative partition, such as predicting the specific outcome {X}, the content is higher than a less informative partition such as {X or Y}.[2] The content of 'not-H' is the content of the complement of the partition, i.e. the disjunction of every other outcome. By subtracting content of the complement of the partition, vaguer partitions take a larger hit to their

[2] For the sake of space, I will leave out the specific mechanics of how content is calculated since it is not relevant to this paper.

EU since their complements have high content. The value of ‘q’ determines the magnitude of this hit in comparison to the effect of ‘p(H)’. Thus, a higher value of ‘q’ will benefit more specific partitions when calculating EU and ultimately when deciding which partition is maximal.

II. Betting Mechanics and Calculating Odds

As a tool for this investigation, I will construct a very simplified model of a sportsbook where bets on horse races would be placed. The realism of the sportsbook is not exceedingly important to my investigation; it serves more to put probabilities and payouts in a less abstract and more easily discussable context. I will refer to the party setting the prices as the House and the decision-making agent placing bets on the House’s sportsbook as the Player.

The potential payout for a bet on an outcome is the multiplicative inverse of the probability at which the outcome is expected to occur. If a bettor places a $1 bet on an outcome with a 25% probability, she will win a $4 payout minus her initial $1 if the outcome occurs and simply lose her $1 if it does not.

These probabilities are set based on a number of factors. In the case of horse race betting, the House (and perhaps the Player too) may initially set the odds according to historical data on the competing horses by using this data to get a sense of which horses can be expected to perform well or poorly. More importantly, demand is the major determinant in pricing bets. If many bettors are betting on a certain outcome, the price of making the bet will increase or the payout associated with that outcome will be lessened. Less popular bets have the potential to yield the highest payouts, because if many bets are made in the opposite direction, the pool of available money is large. By doing this, public impressions are factored into pricing and payout models. As a result, if some piece of evidence that recommends a certain bet is widely dispersed

among bettors, the upside of making that recommended bet diminishes as additional people jump on the metaphorical bandwagon.[3] Much like how stock market prices quickly fluctuate after companies publish quarterly reports and information disperses among investors, a bettor must also be fast or go against the grain to beat the betting market and make a profit. Note that the odds set by the House depend on the bets placed by the collective bettors. Thus, when I refer to the probabilities, odds, or credences of the House, I am referring to the odds the House offers to the Player once they have factored in the demand of the collective bettors.[4]

It usually costs money to place a bet and thus if the bets average out as they are designed to, the Player loses this fee they paid and the House makes a profit as the facilitator of the bets (the House is not meant to make a profit by misleading the public about odds). In my model I will disregard any margin taken by the House as the price of making the bet. That is, there is no fee associated with placing a bet, unlike in the case of real bets on horse races. This assumption serves to simplify the problem so that premiums can be calculated more simply, however the simplification comes at the cost of some degree of realism.

III. Applying Levi's Framework to Toy Scenarios

For the remainder of this paper, I will investigate the effect of varying Levi's parameter 'q' on a varied set of betting scenarios. These scenarios represent different circumstances that might call for different optimal values of 'q'. The overall question is as follows: when and how do particular 'q' values optimize expected winnings? I will begin with a simple case where the

[3] Sohail, Shehryar. "Understand the Math Behind Betting Odds in Gambling." *Investopedia*, Dotdash Meredith, 23 Sept. 2024, www.investopedia.com/articles/dictionary/042215/understand-math-behind-betting-odds-gambling.asp.
[4] Thus, the odds generated by the House reflect the credences or expectations of all the stakeholders; they are not set by one individual or a small group. In fact, the House credences do not necessarily represent the credences of any individual. Rather they represent the average credences among all bettors.

Player and House are in agreement over their credences[5] and examine the effect of varying 'q' for two different distributions of credences: one where win probabilities are evenly dispersed among the horses and one where there is a clear winner. Next, I will examine cases where the Player's credences differ from the one's used to set the odds (the House's credences). I will look at a case where the House's odds are accurate to the true probabilities and the opposite case where the Player's odds are accurate to the true probabilities. Finally, I will look at a case where highly specific bets are assigned premiums to make them more attractive to betters and examine the effect of varying 'q' there.

IV. Case 1a: Universal Agreement

To illustrate the model's functionality, I will begin by examining a toy case where the Player and the House agree exactly on the odds of the race, and their predictions are correct on average. Thus, we expect that the player cannot outperform the House on a consistent basis since their betting odds are identical. While the player might beat the House some of the time, on average they will lose money.

Evenly Dispersed Variation

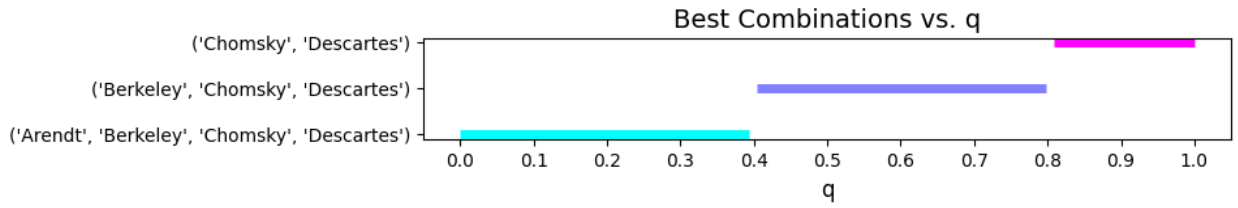
Horses = {Arendt, Berkeley, Chomsky, Descartes}

P = {0.1, 0.2, 0.3, 0.4}[6]

[5] Credences refer to the probabilities of each horse winning.
[6] The four probabilities match respectively to the four horse names.

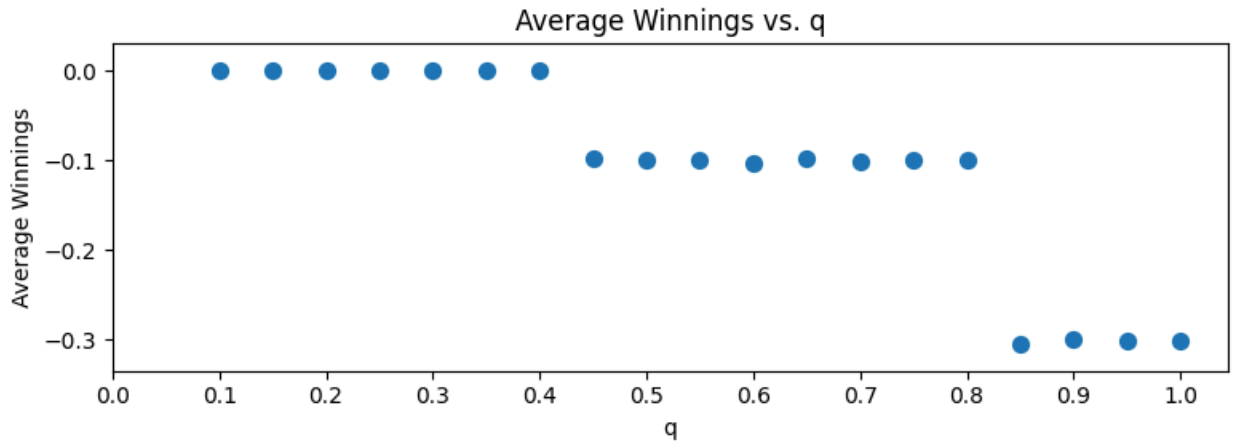
Winning Partition	Odds
[Arendt]	10.0
[Berkeley]	5.00
[Chomsky]	3.33
[Descartes]	2.50
[Arendt, Berkeley]	3.33
[Arendt, Chomsky]	2.50
[Arendt, Descartes]	2.00
[Berkeley, Chomsky]	2.00
[Berkeley, Descartes]	1.67
[Chomsky, Descartes]	1.43
[Arendt, Berkeley, Chomsky]	1.67
[Arendt, Berkeley, Descartes]	1.43
[Arendt, Chomsky, Descartes]	1.25
[Berkeley, Chomsky, Descartes]	1.11
[Arendt, Berkeley, Chomsky, Descartes]	1.00

The above table shows the odds associated with each partition. The odds are generated by the House's credences,[7] which in this case are identical to the Player's. As explained previously, these odds correspond to the payout the Player would receive if they were to correctly place a $1 bet on a horse contained in their partition winning. For the maximally vague bet that includes all four horses as possible winners, the payout is exactly $1, demonstrating that no money can be earned from betting on that outcome since it is guaranteed. In this feeless system, a maximally vague bet is essentially the same as making no bet at all.

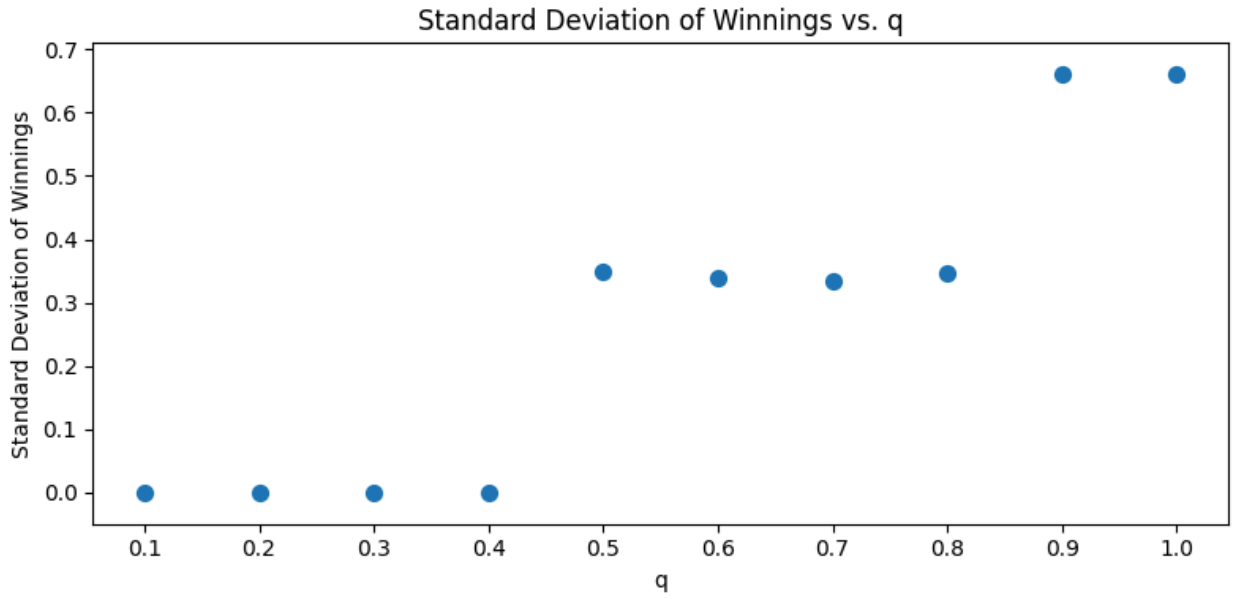


The above graph demonstrates how the partition with the maximal EU varies depending on 'q'. In other words, as the agent is more inclined to value relief from agnosticism over avoidance of error, the partition they select for their bet based on EU changes at certain thresholds. As expected, the size of the optimal partitions mirror the increasing 'q' values by becoming more and more specific. For every $q < 0.4$, EU is maximized by making the maximally vague bet. As mentioned before, this translates to not making a bet at all. Thus, when $q < 0.4$, no bet is considered worth the risk of erring.

[7] This is simply because the House regulates the bets. Additionally, they are responsible for aggregating the credences expressed by every bet purchased, and minimize their own volatility by ensuring that every bet has someone on both sides.



In calculating the Average Expected Winnings associated with a certain ‘q’ value, the race is simulated according to a set of true credences that are not necessarily the same as either the House or Player’s credences. The Player makes a prediction on the partition with the maximal EU according to their credences, Levi’s framework, and the designated ‘q’ value, and then the race is simulated 10,000 times based on the true probabilities. The Average Expected Winnings for the race is the Player’s average winnings across these 10,000 simulations, assuming they get the corresponding payout based on the House’s odds if they are correct, and lose their $1 stake if not. Here, it is evident that the Player never breaks even and becomes less successful as their bets become more specific.. This aligns with our expectations: they have no upper hand since their credences are identical to those of the House. Minor deviations from this pattern are mere noise; no bet is optimal since each is expected to result in zero or negative winnings.



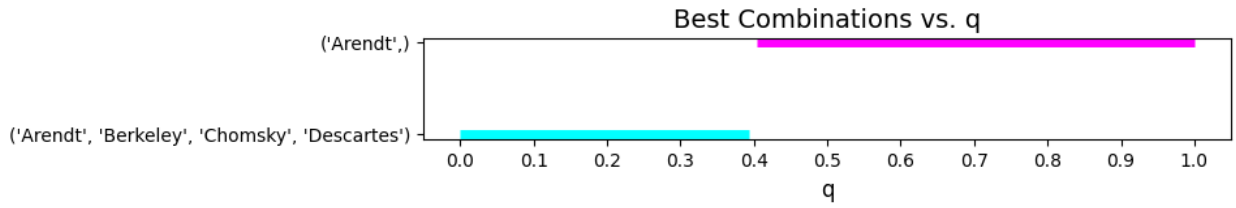
The above graph demonstrates the increased risk associated with making more specific bets. Although Average Expected Winnings was calculated based on the average of many trials, it is important to note that specific bets are more volatile than vague ones. In reality, bets on horse races are not made thousands of times such that the Average Expected Winnings associated with a bet are always guaranteed, and more specific bets are less successful on average. They are made worthwhile due to the higher reward potentially received. Taking on this extra risk makes specific bets less appealing, although their greater payout might justify their volatility to risk-seeking bettors.

In general, this first case provides confirmation that the model is working as we expect it to. These results indicate that when the Player agrees with the House on the probabilities of every partition, these probabilities are exactly accurate to the average performance of the horses, and payouts correspond exactly to the reciprocal of the outcome probabilities, the Player has zero chance of positive earnings.

V. Case 1b : Universal Agreement with a Clear Winner

Horses = {Arendt, Berkeley, Chomsky, Descartes}

P = {0.7, 0.1, 0.1, 0.1}



Briefly note this case where one horse has a much higher chance of winning than the others.[8] Here, a bet on only Arendt becomes preferable at q = 0.4, whereas in the previous case a bet on a specific horse never had the maximal EU.

VI. Differing Credences

Placing bets on horse races would not be a worthwhile endeavor if the player did not have any chance of making money. So far, we have only examined cases where the House and the Player are in complete agreement about what chances each horse has of winning. Moving past Case 1, the more interesting bets that result in earnings will involve differing credences between the House and the Player.

There are two ways in which this divergence in credences could arise. First, the Player and the House might interpret the same body of evidence differently. For example, say the Player and the House, representing the average of all other bettors, all know that Chomsky is racing with a new jockey in this race. Perhaps the House is worried about this negatively affecting the

[8] According to the Player, the House, and the true probabilities.

horse's performance, but the player thinks it will make no difference. This could result in differing credences between the House and the Player. Second, the Player and the House may operate under information asymmetry, basing their credences on different bodies of evidence. The Player could bribe someone to sabotage a certain horse and then bet against that horse based on their secret misdeed. Information asymmetry could also take more idiosyncratic forms. Maybe the Player happens to catch a glimpse of Berkeley's jockey looking anxious before the race. Perhaps the Player had a dream the night before that Arendt would win the race and places their bet based off of this. The separation between what is included in the public body of evidence can be drawn in many ways, and there could be disagreement or overlap between what counts as new evidence versus interpretation of existing evidence or evidence with debatable validity.

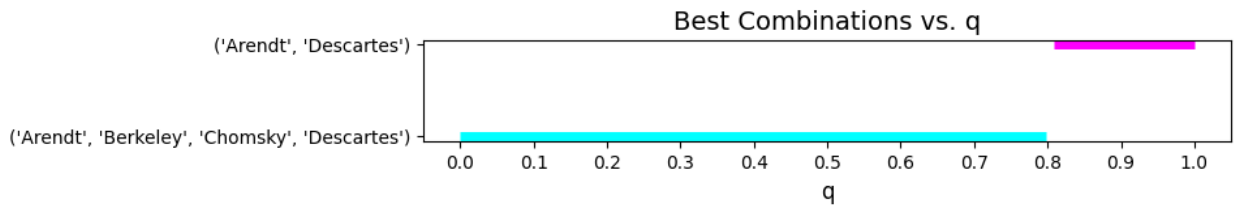
VII. Case 2a: The House Has Correct Credences

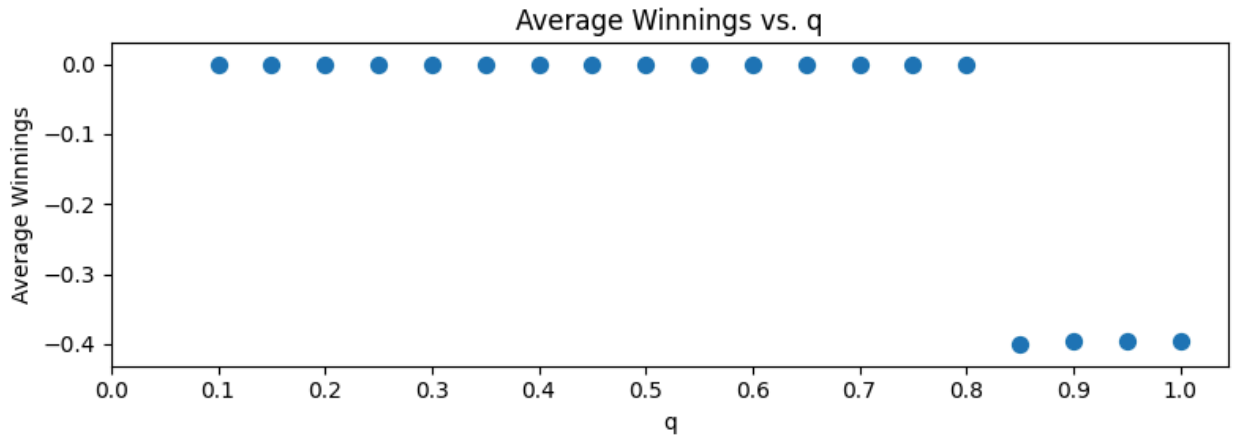
Horses = {Arendt, Berkeley, Chomsky, Descartes}

P_{actual} = {0.5, 0.2, 0.2, 0.1}

P_{Player} = {0.3, 0.2, 0.2, 0.3}

P_{House} = {0.5, 0.2, 0.2, 0.1}





In this case, we see the expected result that holding faulty credences causes an increase in specificity to be punished. This is intuitively obvious. By increasing in specificity, the Player exposes themself to more error, a risk that is exacerbated by the fact that this specificity zooms further in on a faulty basis of credences.

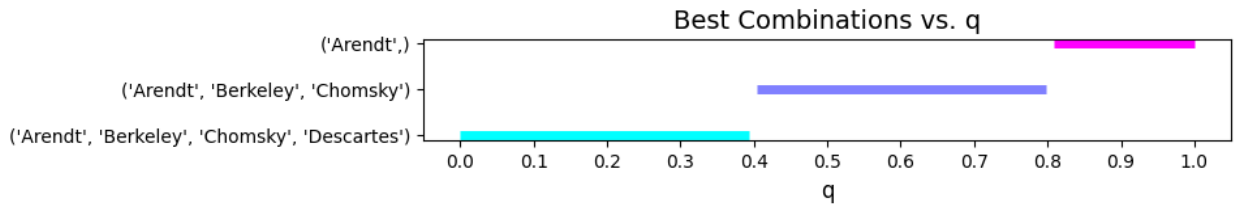
VIII. Case 2b: The Player Has Correct Credences

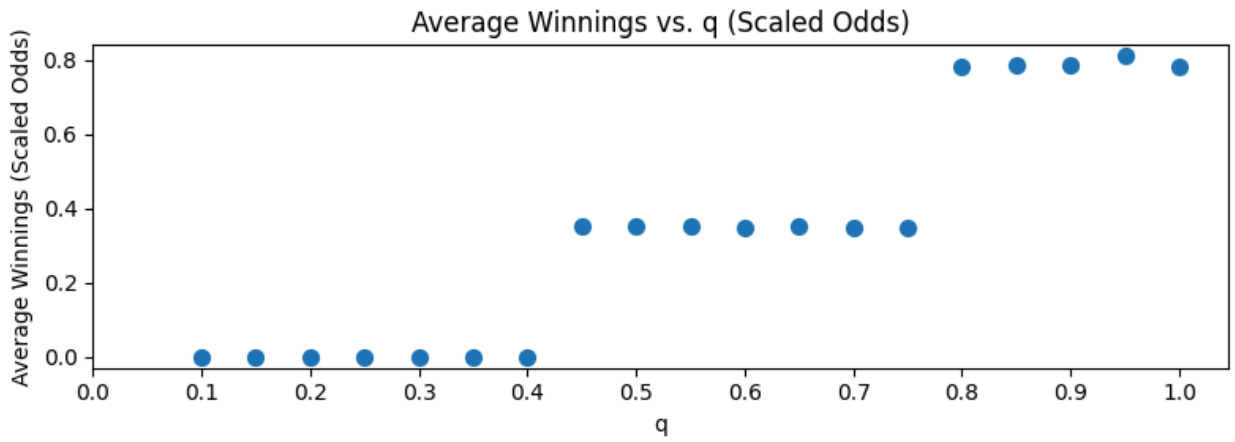
Horses = {Arendt, Berkeley, Chomsky, Descartes}

P_{actual} = {0.5, 0.2, 0.2, 0.1}

P_{Player} = {0.5, 0.2, 0.2, 0.1}

P_{House} = {0.3, 0.2, 0.2, 0.3}





In the alternative case where the Player is correct about the true probabilities and the House is mistaken, we finally see a way in which an increased 'q' value leads to earnings. This case is what bettors presumably strive for; through either a unique interpretation of public information or the use of private information, the bettor outplays the House and earns back more than their original bet.

IX. Specificity Premiums

In the earlier discussion of the volatility associated with more specific bets in Case 1a,[9] we noted that specific bets are rewarded with higher payouts in order to compensate for the risk being taken. In practice, premiums are paid to risk-takers in many different forms. A risky job such as underwater welding comes with higher pay. An investment into an asset manager with higher volatility comes with the expectation that returns will be higher on average. Risk can disappear from view not in our simulations where many trials are run and averages converge to a singular value. However, it is an important factor for bettors to consider since they are not

[9] See page 10.

repeating their bet over the course of many trials. Since investors and bettors seek to minimize risk, an oversized payout is commonly expected for making these bets. This section will examine the effect of varying the value of 'q' in the special case where the House sets payouts such that they reward riskier bets.

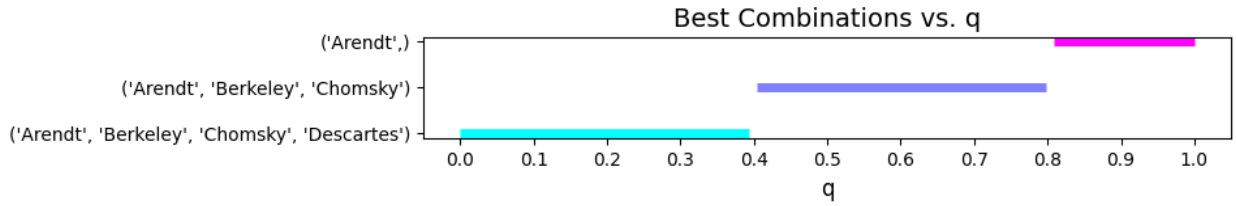
Let us return to the cases of the driver facing an imminent crash and the adventurous foodie. The difference between these two cases is in the stress of their utilities on specificity or vagueness. In the driving case, the driver faces extremely negative utility if they make a vague decision (i.e. failing to choose what to do). The foodie, on the other hand, could potentially have extremely negative utility upon making a specific decision and is mostly fine in the case they fail to make a specific decision and relieve their agnosticism. The placement of the extreme negative on being either overly vague or overly risk-seeing represent the bigger picture of when specificity or vagueness are rewarded (or punished) with outsized utilities. The specificity premium for bets in this case demonstrates how strategic placement of utility can be used to encourage certain preferences in decision-making.

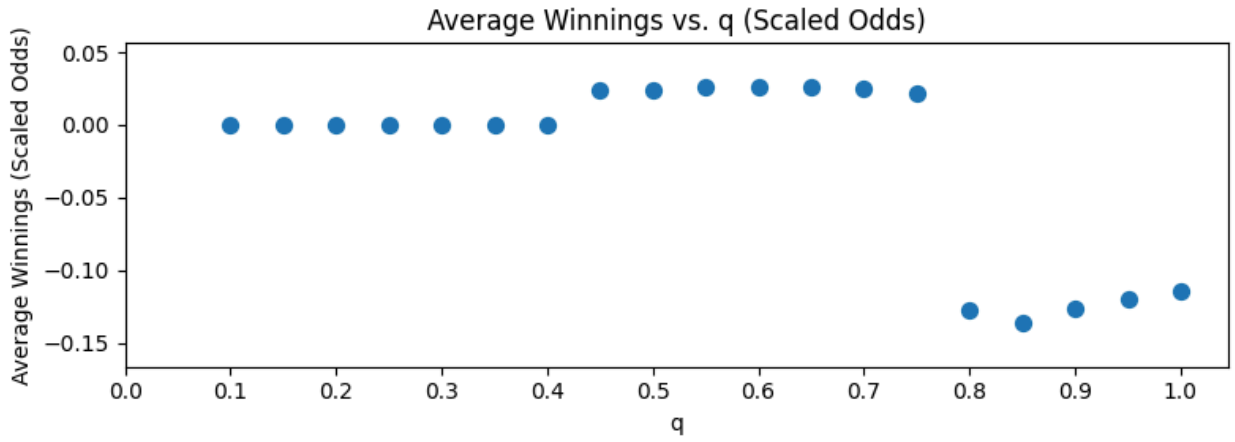
X. Case 3a: Rewards for Specific Bets, Universal Agreement

Let us return to Case 1b with evenly dispersed probabilities among the horses and universal agreement between the House, the Player, and the true probabilities. However, the payouts have been scaled such that bets have higher rewards when their partitions have higher content. In this case, each payout receives a bonus based on the content of its partition as a percentage divided by two. For example, a bet on a partition with a singular element would be increased by (¾)/2%, or 37.5%. The amount and mechanics are arbitrary; the important change is that the more specific a bet is, the more its payout increases based on this scaling.

Horses = {Arendt, Berkeley, Chomsky, Descartes}

P = {0.1, 0.2, 0.3, 0.4}





In this case of universal agreement on probabilities, the higher ‘q’ values lead the Agent to losses since the times Arendt happened to lose still outweigh the benefits of the specificity premium. However, a slight benefit from the specificity premium is evident for values of ‘q’ falling between 0.45 and 0.75. Evidently, given this probability distribution, a slightly more specific bet is now preferable to making no bet at all. Compare this to the earlier example 1a where all bets other than the maximally vague bet resulted in a net loss. However, the payouts have not been scaled enough that specificity is infinitely beneficial. Once ‘q’ is high enough that the chosen partition contains a singular element, the specificity premium is no longer worth the increased chance of error.

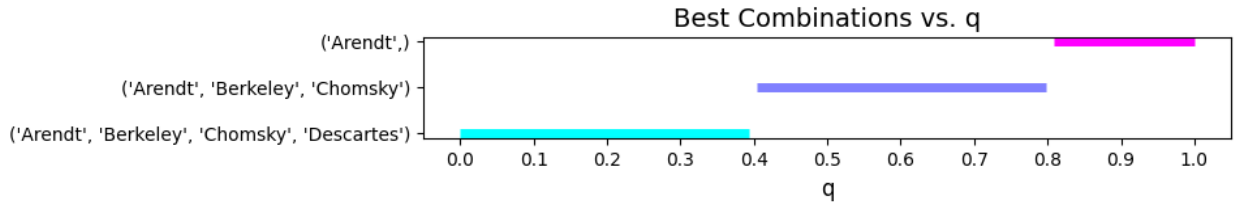
XI. Case 3b: Rewards for Specific Bets, The Player Has Correct Credences

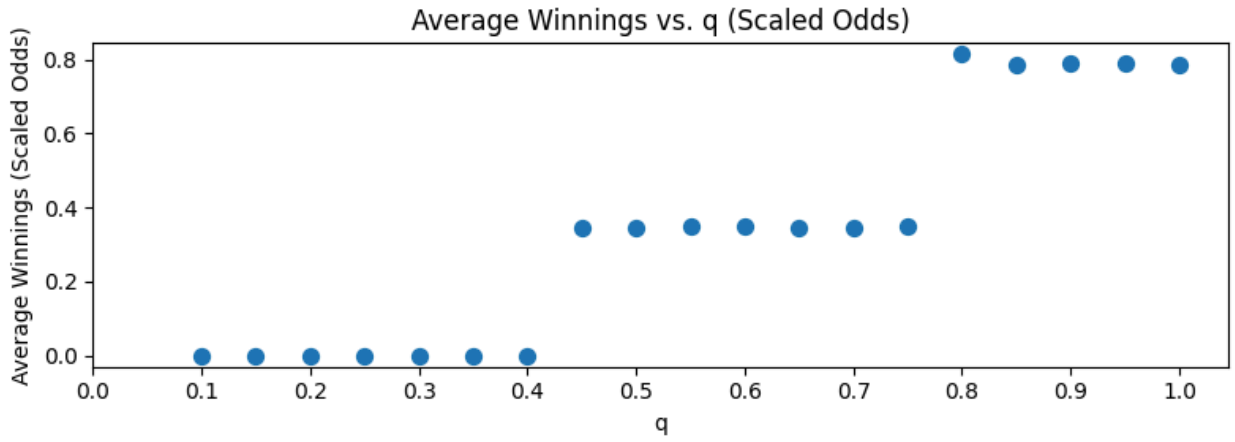
Horses = {Arendt, Berkeley, Chomsky, Descartes}

P_{actual} = {0.5, 0.2, 0.2, 0.1}

P_{Player} = {0.5, 0.2, 0.2, 0.1}

P_{House} = {0.3, 0.2, 0.2, 0.3}





This final case demonstrates the reward for specificity in a successful risky bet in a sportsbook that rewards risk-takers with a premium. Not only has the Agent made a bet based on more accurate probabilities than the House, but they can also benefit from doubling down on their credences to make a highly specific bet and reap the premium.

XII. Conclusion

While this paper presents the preliminary steps of assessing the effect of varying ‘q’ in Levi’s model, there are many ways in which the investigation could be taken further. One underexplored factor in this assessment of ‘q’ values is the effect of the exact probability distributions on the way Levi’s framework selects the optimal partition. This was demonstrated in the comparison between Cases 1a and 1b, but the exact way by which the probabilities stand in relation to each other would be an interesting topic for further investigation. Additionally, varying the size and calculation method of the specificity premium from Cases 3a and 3b could be an important factor to look at to understand whether our results are generalizable or rely on this particular way of determining the size of the premium. Expanding this analysis could offer additional insights into the applicability of Levi’s approach under different conditions. This further exploration would not only deepen our understanding of his model but also help refine its applications to practical decision problems.